Industrial Internship Report on

” Quality Prediction in a Mining Process”

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Executive Summary

This report presents the work completed during the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. (UCT).

The project, titled Quality Prediction in a Mining Process, focuses on predicting the percentage of silica concentrate in an iron ore flotation process using machine learning techniques.

The project involved data preprocessing, exploratory data analysis, feature analysis, model development, and performance evaluation. Random Forest and Extra Trees Regressor algorithms were implemented and compared to identify a suitable prediction model.

The Extra Trees Regressor achieved an MAE of 0.5726, an RMSE of 0.7157, and an R^2 score of 0.6100 on the validation dataset.

This internship provided practical experience in data analysis, machine learning implementation, model comparison, and applying predictive techniques to an industrial process.

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1 Preface

The six-week Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. (UCT) was a valuable learning experience.

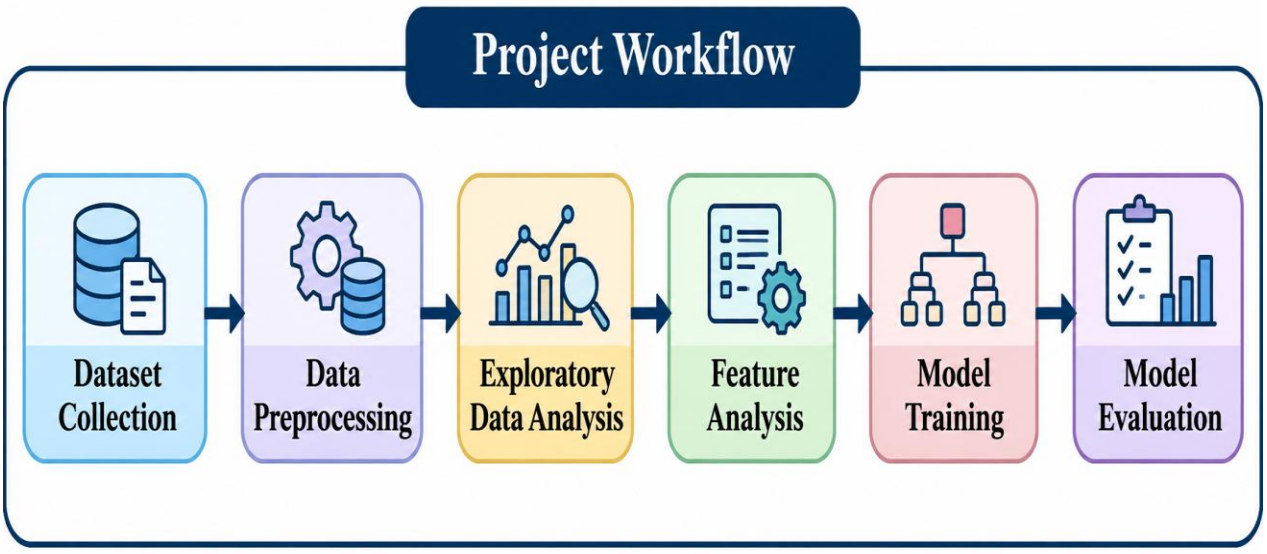
During this internship, I worked on Quality Prediction in a Mining Process, a machine learning project focused on predicting silica concentrate quality in an iron ore flotation process using historical industrial data.

The project involved data preprocessing, exploratory data analysis, feature analysis, Random Forest and Extra Trees model development, performance evaluation, and result comparison.

The internship provided practical exposure to real-world industrial data and helped me improve my technical knowledge, analytical thinking, and machine learning skills.

I sincerely thank upskill Campus, The IoT Academy, UCT, and everyone who supported me throughout this internship.

I encourage my juniors and peers to participate in such programs to gain practical experience and develop industry-relevant skills.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

• Build Your own dashboard

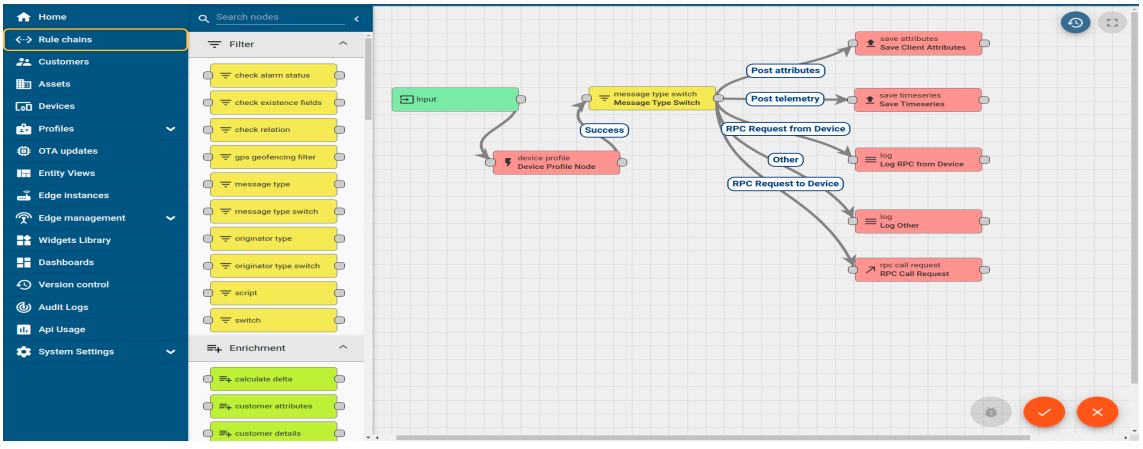
• Analytics and Reporting

• Alert and Notification

• Integration with third party application(Power BI, SAP, ERP)

• Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

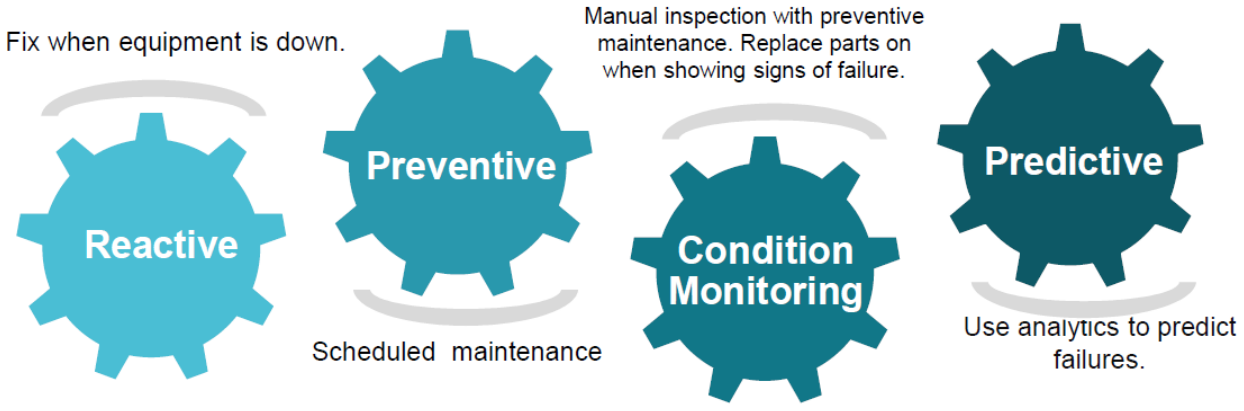


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

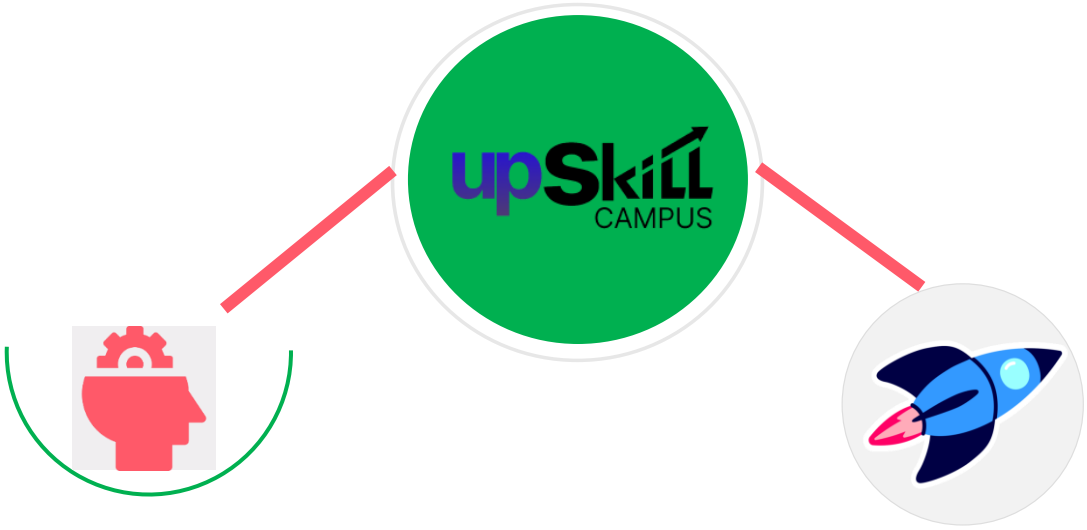
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

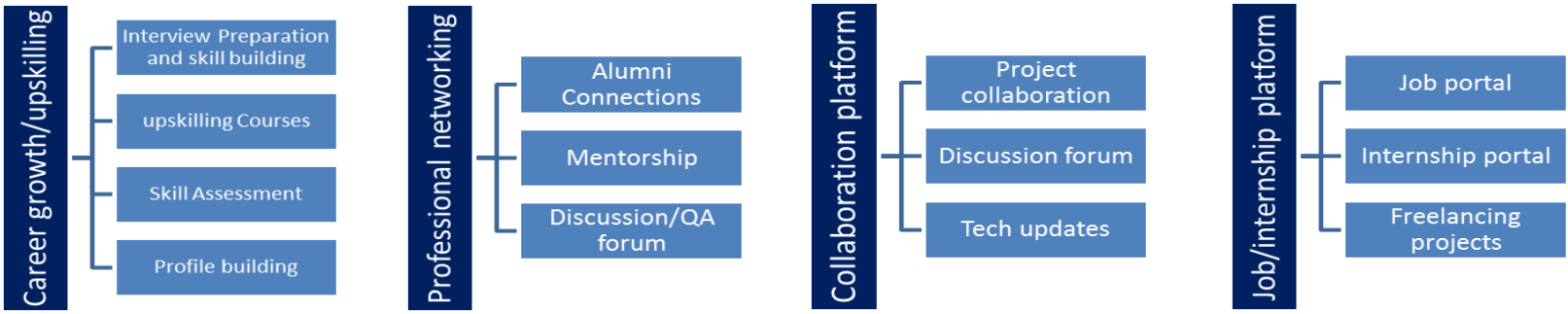
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The main objectives of this internship program were:

- ☛ To gain practical experience in solving real-world industrial problems.
- ☛ To understand the application of machine learning in industrial processes.
- ☛ To develop skills in data preprocessing, analysis, and visualization.
- ☛ To implement and compare machine learning regression algorithms.
- ☛ To evaluate model performance using suitable evaluation metrics.
- ☛ To improve analytical thinking, technical knowledge, and problem-solving abilities.

2.5 Reference

[1] Scikit-learn Documentation – Random Forest Regressor. https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html

[2] Scikit-learn Documentation – Extra Trees Regressor. https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.ExtraTreesRegressor.html

[3] Pandas Documentation. https://pandas.pydata.org/docs/

[4] NumPy Documentation. https://numpy.org/doc/

[5] Matplotlib Documentation. https://matplotlib.org/stable/

2.6 Glossary

Term	Acronym / Meaning
Machine Learning	ML
Exploratory Data Analysis	EDA
Random Forest	RF
Extra Trees Regressor	ET
Mean Absolute Error	MAE
Root Mean Squared Error	RMSE
Coefficient of Determination	R^2
Silica Concentrate	Percentage of silica in concentrate
Iron Concentrate	Percentage of iron in concentrate
Feature Importance	Contribution of a feature to model prediction

3 Problem Statement

The mining industry involves complex processes in which maintaining the quality of the final product is essential. In iron ore flotation, the percentage of silica concentrate is an important quality indicator that affects the purity of the final iron concentrate.

Traditional methods of monitoring and controlling the flotation process may not always identify complex relationships between process parameters and product quality.

The objective of this project, **Quality Prediction in a Mining Process**, is to develop a machine learning model that predicts the percentage of silica concentrate using historical industrial process data.

The project aims to:

1. Analyze historical data collected from the iron ore flotation process.
2. Understand the relationship between process parameters and silica concentrate quality.
3. Identify important features that influence the prediction.
4. Develop machine learning regression models for quality prediction.
5. Evaluate and compare model performance using suitable metrics.

4 Existing and Proposed solution

4.1 Existing Solution

Traditional quality monitoring in mining processes relies on laboratory analysis and process measurements to assess the quality of the final concentrate.

However, these methods may have limitations:

- Laboratory testing may require additional time.
- Multiple process parameters make quality analysis complex.
- Identifying relationships between process variables and product quality can be difficult.
- Manual analysis may not provide sufficient predictive insights.

4.2 Proposed Solution

The proposed system uses machine learning regression algorithms to predict the percentage of silica concentrate based on historical iron ore flotation process data.

Random Forest and Extra Trees Regressor algorithms were implemented and evaluated to identify a suitable prediction model.

The proposed approach includes:

1. Collecting and loading historical mining process data.
2. Cleaning and preprocessing the dataset.
3. Performing exploratory data analysis.
4. Identifying important process parameters using feature importance.
5. Training Random Forest and Extra Trees regression models.
6. Evaluating and comparing model performance using MAE, RMSE, and R^2 score.

4.3 Code Submission (GitHub Link)

GitHub Repository: ***https://github.com/nikitamali21/upskillCampus/blob/main/Project-10-Mining-Quality-Prediction/Project10_Mining_Quality.ipynb***

4.4 Report Submission (GitHub Link)

Project Report: ***https://github.com/nikitamali21/upskillCampus/blob/main/Project-10-Mining-Quality-Prediction/MiningProcessQualityPrediction_Nikita_USC_UCT.pdf***

5 Proposed Design/ Model

The system processes historical industrial data, identifies important process parameters, trains regression models, and evaluates their performance. Random Forest and Extra Trees Regressor models are compared to determine the better-performing model.

5.1 High Level Diagram

This diagram represents the complete workflow, from loading the mining dataset to predicting silica concentrate quality.



Figure5. 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

The detailed implementation consists of the following steps:

1. Data Loading: Load the mining process dataset using Pandas.
2. Data Preprocessing: Convert comma-separated decimal values into numerical format, remove duplicate records, and check for missing values.
3. Exploratory Data Analysis: Analyze the distribution of process parameters and silica concentrate quality using statistical summaries and visualizations.
4. Feature Analysis: Identify important input parameters and study their contribution to silica concentrate prediction.
5. Model Training: Train Random Forest and Extra Trees Regressor models using the selected process parameters.
6. Model Evaluation: Compare both models using MAE, RMSE, and R^2 score.
7. Prediction: Use the selected model to predict silica concentrate percentage.

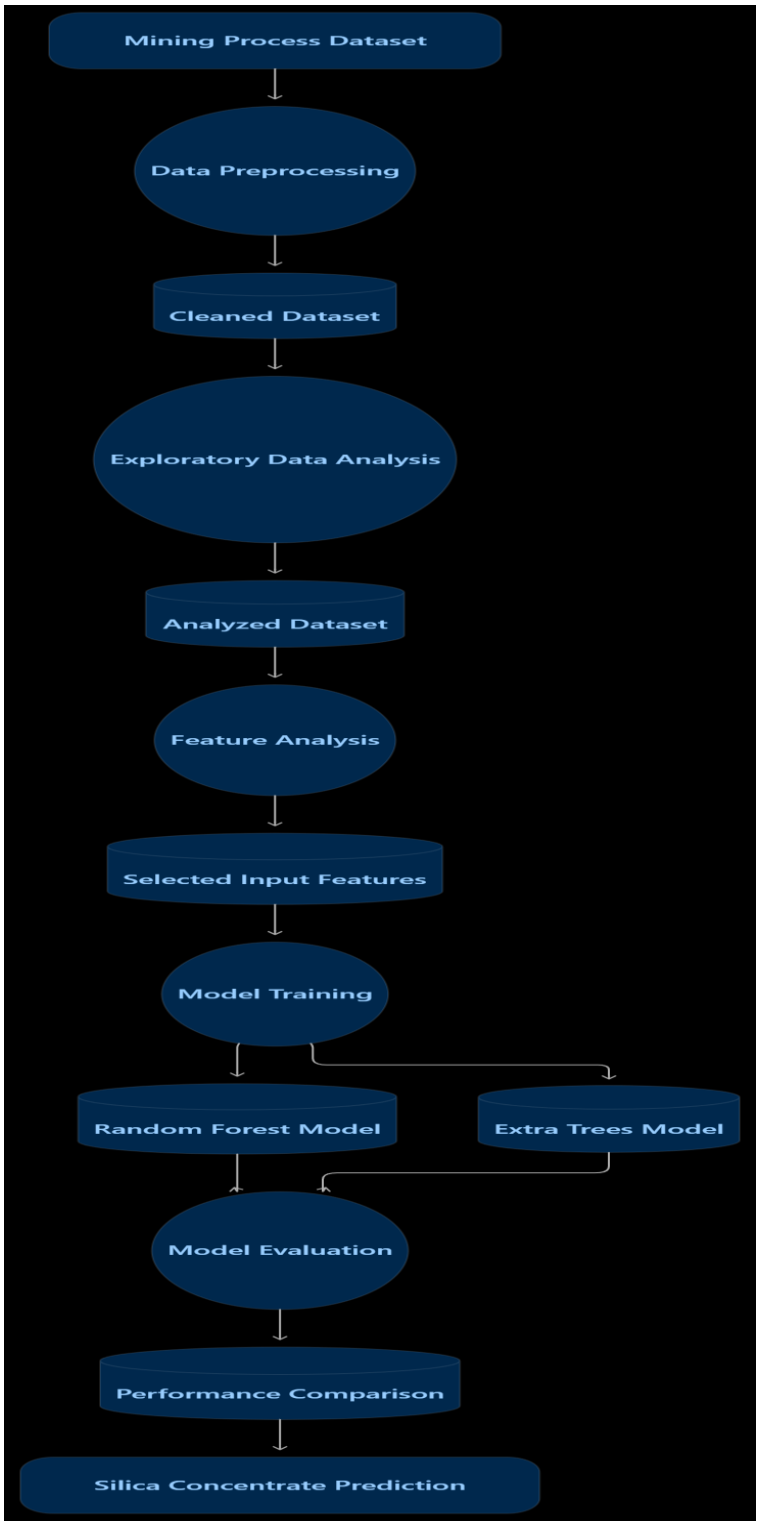


Figure 5.2: Data Flow Diagram (DFD)

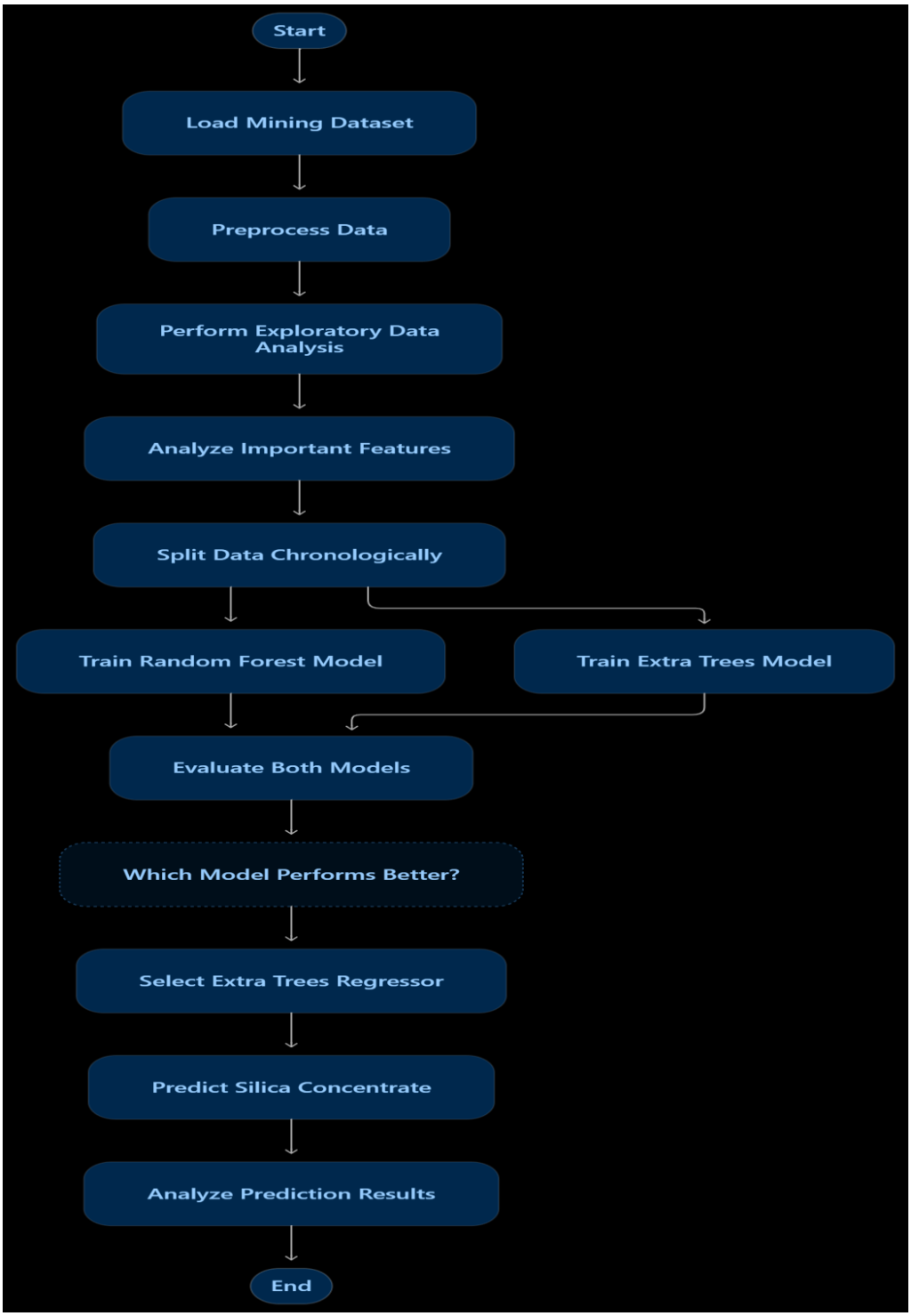


Figure 5.3: Flowchart

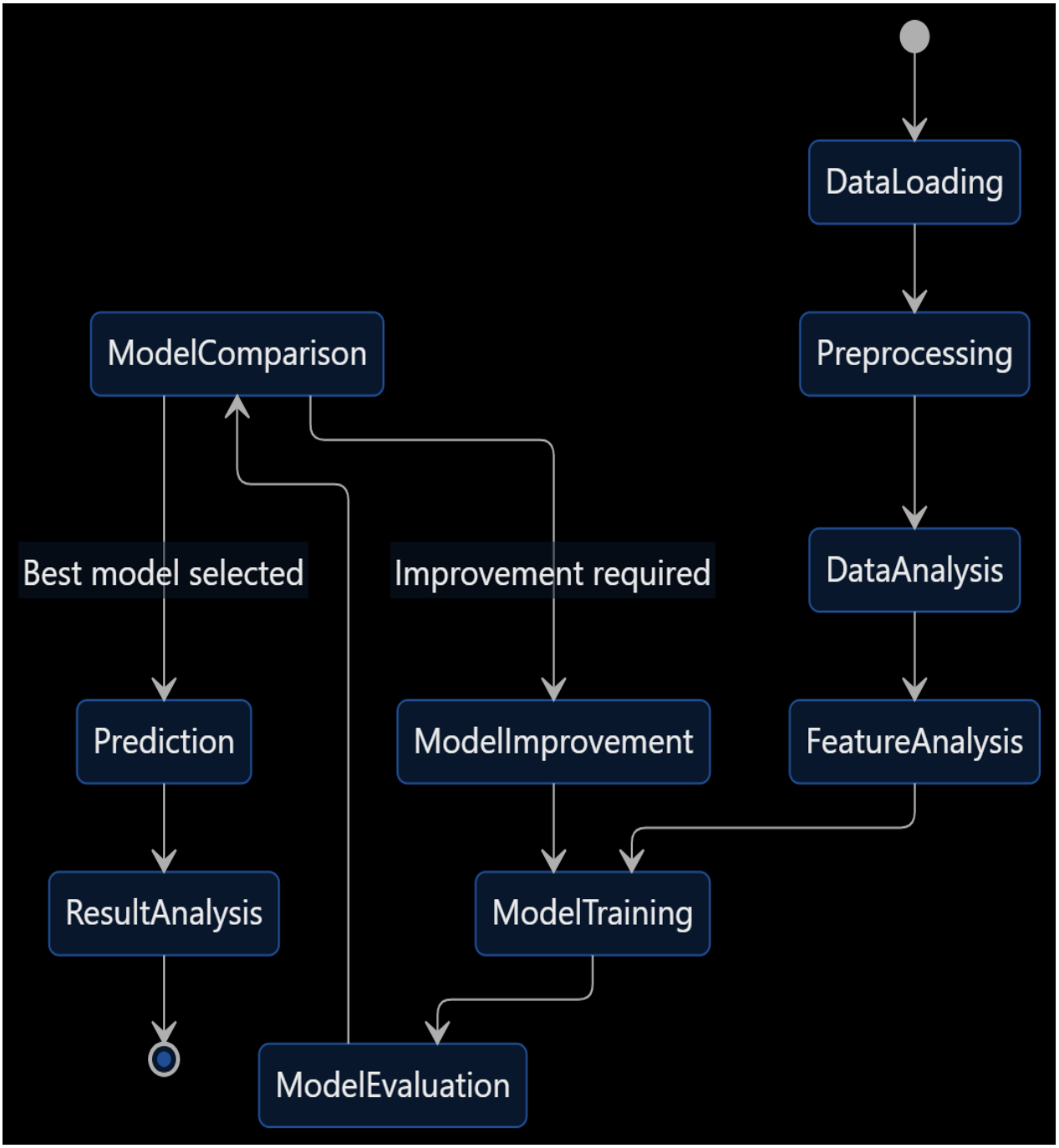


Figure 5.4: State Machine

5.3 Interfaces (if applicable)

The project was implemented using Python in a notebook-based environment. The main tools and libraries used were:

- Google Colab / Jupyter Notebook: For writing and executing Python code.
- Pandas: For loading, cleaning, and manipulating the mining dataset.
- NumPy: For numerical operations.
- Matplotlib and Seaborn: For data visualization and exploratory analysis.
- Scikit-learn: For implementing Random Forest and Extra Trees Regressor models and evaluating their performance.
- CSV Files: For storing the mining dataset and prediction results.

```
# Load the original mining flotation dataset

mining_raw = pd.read_csv("/content/MiningProcess_Flotation_Plant_Database.csv")

print("Dataset Shape:", mining_raw.shape)

print("\nDate range:")
print(mining_raw['date'].min(), "to", mining_raw['date'].max())

print("\nFirst 5 rows:")
display(mining_raw.head())
```

Dataset loading and preprocessing.

```
# Check target statistics and time range

print("Date range:")
print(mining_clean['date'].min(), "to", mining_clean['date'].max())

print("\nUnique timestamps:")
print(mining_clean['date'].nunique())

print("\nTarget: % Silica Concentrate")
print(mining_clean['% Silica Concentrate'].describe())

print("\nTarget missing values:")
print(mining_clean['% Silica Concentrate'].isnull().sum())
```

```
Date range:
2017-03-10 01:00:00 to 2017-09-09 23:00:00

Unique timestamps:
4097

Target: % Silica Concentrate
count    736282.000000
mean          2.327270
std           1.125616
min           0.600000
25%           1.440000
50%           2.000000
75%           3.010000
max           5.530000
Name: % Silica Concentrate, dtype: float64

Target missing values:
0
```

Exploratory data analysis and visualizations.

```
print("Available Project 10 analysis variables:\n")

for name, obj in globals().copy().items():
    if any(word in name.lower() for word in [
        'residual', 'error', 'actual', 'pred', 'daily', 'comparison'
    ]):
        if not name.startswith('_'):
            print(f"{name}: {type(obj).__name__}")
```

```
Available Project 10 analysis variables:

mean_absolute_error: function
mean_squared_error: function
y_pred_baseline: ndarray
y_pred_no_iron: ndarray
actual_mining: ndarray
predicted_mining: ndarray
residuals_mining: ndarray
error_analysis_mining: DataFrame
largest_errors_mining: DataFrame
daily_errors: DataFrame
comparison_mining: DataFrame
y_pred_extra: ndarray
final_comparison_mining: DataFrame
residuals_extra: ndarray
comparison_check: DataFrame
```

Feature Importance analysis

```
# Baseline Random Forest

rf_baseline = RandomForestRegressor(
    n_estimators=30,
    max_depth=15,
    random_state=42,
    n_jobs=-1
)

print("Training baseline model...")

rf_baseline.fit(
    X_train_mining,
    y_train_mining
)

print("Baseline model training completed!")
```

```
Training baseline model...
Baseline model training completed!
```

Random Forest training

```python
# Make predictions
y_pred_baseline = rf_baseline.predict(X_val_mining)

# Calculate evaluation metrics
mae_baseline = mean_absolute_error(
    y_val_mining,
    y_pred_baseline
)

rmse_baseline = np.sqrt(
    mean_squared_error(
        y_val_mining,
        y_pred_baseline
    )
)

r2_baseline = r2_score(
    y_val_mining,
    y_pred_baseline
)

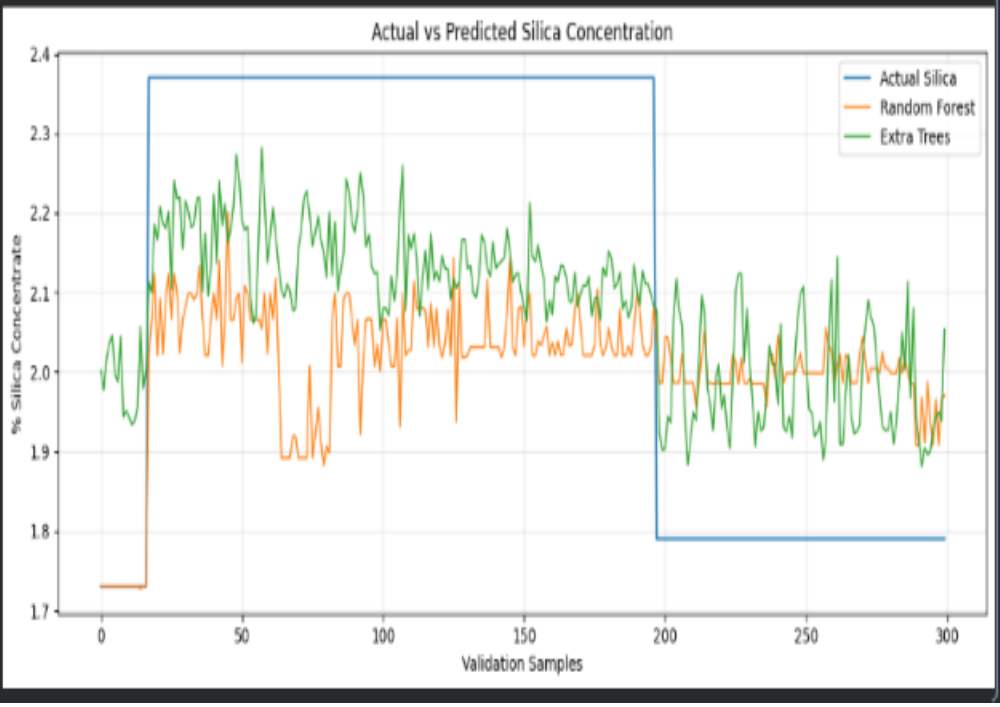
print("Baseline Random Forest Results")
print("--------------------------------")
print("MAE :", round(mae_baseline, 4))
print("RMSE:", round(rmse_baseline, 4))
print("R²  :", round(r2_baseline, 4))
```

```
Baseline Random Forest Results
--------------------------------
MAE : 0.6507
RMSE: 0.8254
R²  : 0.4813
```

Model evaluation and performance comparison

```
plt.figure(figsize=(11, 5))

plt.plot(
    np.asarray(y_val_mining)[:300],
    label='Actual Silica',
    linewidth=1.5)
plt.plot(
    y_pred_baseline[:300],
    label='Random Forest',
    linewidth=1.2)
plt.plot(
    y_pred_extra[:300],
    label='Extra Trees',
    linewidth=1.2)
plt.xlabel("Validation Samples")
plt.ylabel("% Silica Concentrate")
plt.title("Actual vs Predicted Silica Concentration")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



Actual vs. predicted silica concentrate values

6 Performance Test

The performance of the machine learning models was evaluated using a chronological validation dataset. Random Forest and Extra Trees Regressor algorithms were compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

6.1 Test Plan/ Test Cases

The following test cases were considered to verify the functionality and performance of the system:

Test Case	Description	Expected Result
TC01	Load mining process dataset	Dataset loads successfully
TC02	Convert comma-separated decimal values	Values are converted into numerical format
TC03	Remove duplicate records	Duplicate records are removed
TC04	Perform exploratory data analysis	Data patterns and distributions are identified
TC05	Analyze feature importance	Important process parameters are identified
TC06	Train Random Forest model	Model trains successfully
TC07	Train Extra Trees model	Model trains successfully
TC08	Evaluate model performance	MAE, RMSE and R^2 scores are calculated
TC09	Compare model performance	Better-performing model is identified

6.2 Test Procedure

The following procedure was followed to test the system:

1. Loaded the historical mining process dataset.
2. Converted comma-separated decimal values into numerical format.
3. Removed duplicate records and checked for missing values.
4. Performed exploratory data analysis to understand the dataset.
5. Divided the dataset into training and validation sets chronologically.
6. Trained the Random Forest Regressor model.
7. Trained the Extra Trees Regressor model.
8. Evaluated both models using MAE, RMSE, and R^2 score.
9. Compared the results to identify the better-performing model.

6.3 Performance Outcome

Two machine learning regression models were evaluated and compared on the validation dataset.

Evaluation Metric	Random Forest	Extra Trees
MAE	0.6507	0.5726
RMSE	0.8254	0.7157
R^2 Score	0.4813	0.6100

The Extra Trees Regressor achieved better performance than the Random Forest model. It obtained a lower MAE of 0.5726 and RMSE of 0.7157, along with a higher R^2 score of 0.6100.

Feature importance analysis showed that **% Iron Concentrate** was the most influential feature, contributing approximately 74.59% of the model's feature importance.

The results indicate that the Extra Trees Regressor performed better than the Random Forest Regressor for predicting silica concentrate percentage on the validation dataset.

```python
final_comparison_mining = pd.DataFrame({
    'Model': [
        'Random Forest (All Features)',
        'Random Forest (Without Iron)',
        'Extra Trees (All Features)'
    ],
    'MAE': [
        mae_baseline,
        mae_no_iron,
        mae_extra
    ],
    'RMSE': [
        rmse_baseline,
        rmse_no_iron,
        rmse_extra
    ],
    'R2': [
        r2_baseline,
        r2_no_iron,
        r2_extra
    ]
})

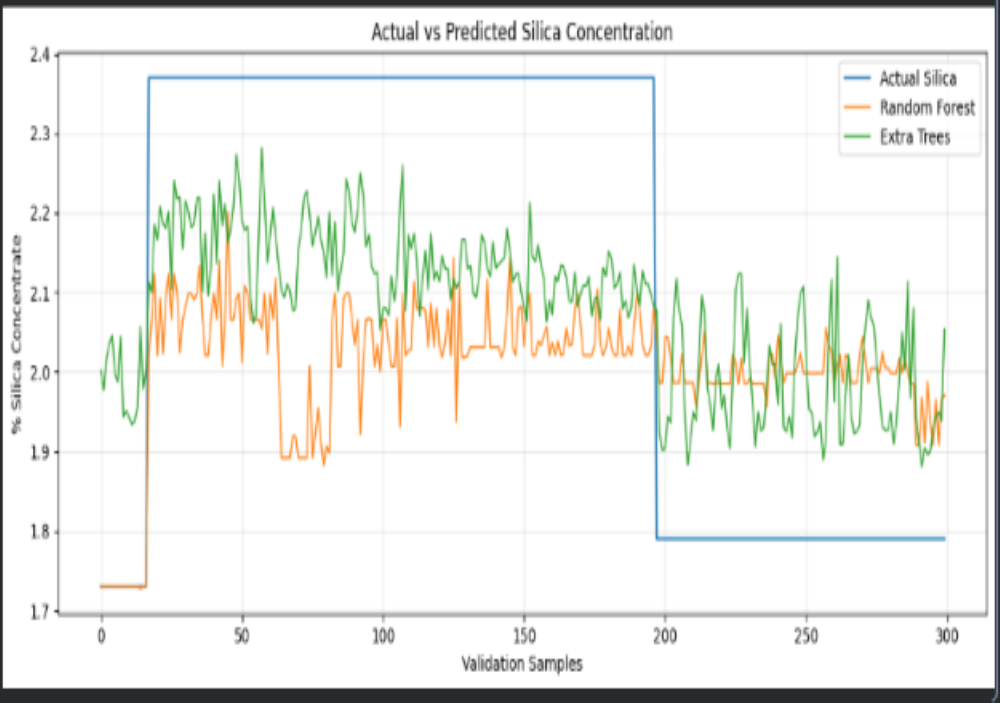
display(
    final_comparison_mining
    .sort_values('MAE')
    .round(4)
    .reset_index(drop=True)
)
```

	Model	MAE	RMSE	R2
0	Extra Trees (All Features)	0.5726	0.7157	0.6100
1	Random Forest (All Features)	0.6507	0.8254	0.4813
2	Random Forest (Without Iron)	0.9804	1.2418	-0.1742

Random Forest vs. Extra Tree performance comparison

```
plt.figure(figsize=(11, 5))

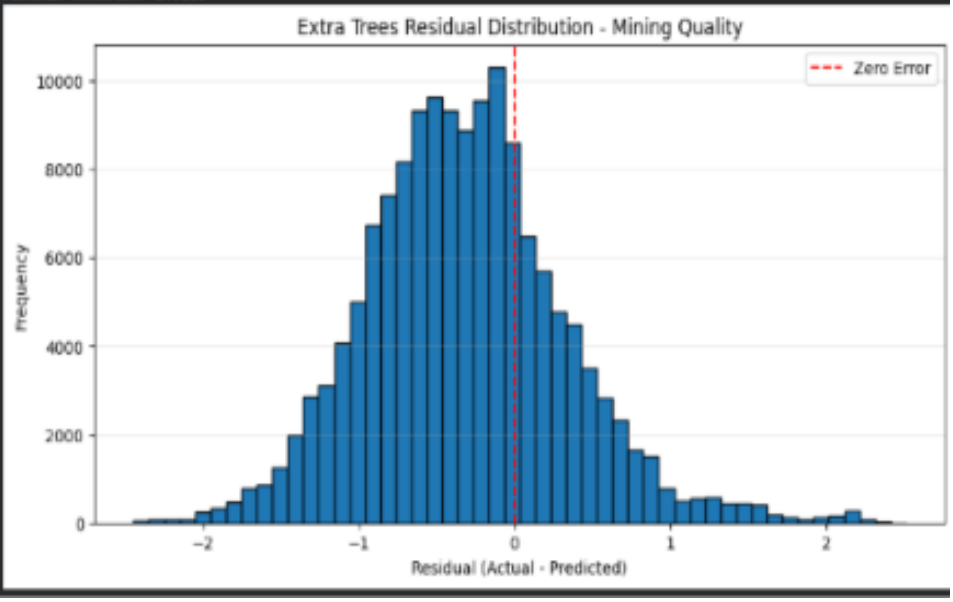
plt.plot(
    np.asarray(y_val_mining)[:300],
    label='Actual Silica',
    linewidth=1.5)
plt.plot(
    y_pred_baseline[:300],
    label='Random Forest',
    linewidth=1.2)
plt.plot(
    y_pred_extra[:300],
    label='Extra Trees',
    linewidth=1.2)
plt.xlabel("Validation Samples")
plt.ylabel("% Silica Concentrate")
plt.title("Actual vs Predicted Silica Concentration")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



Actual vs. Predicted Silica Concentration graph

```
residuals_extra = (
    np.asarray(y_val_mining) -
    np.asarray(y_pred_extra))
print("Extra Trees Residual Analysis")
print("-----------------------------")
print("Mean Residual:",
      round(np.mean(residuals_extra), 4))
print("MAE:",
      round(mean_absolute_error(y_val_mining, y_pred_extra), 4))
print("Maximum Absolute Error:",
      round(np.max(np.abs(residuals_extra)), 4))
print("Minimum Residual:",
      round(np.min(residuals_extra), 4))
print("Maximum Residual:",
      round(np.max(residuals_extra), 4))
plt.figure(figsize=(10, 5))
plt.hist(
    residuals_extra,
    bins=50,
    edgecolor='black')
plt.axvline(
    0,
    color='red',
    linestyle='--',
    label='Zero Error')
plt.xlabel("Residual (Actual - Predicted)")
plt.ylabel("Frequency")
plt.title("Extra Trees Residual Distribution - Mining Quality")
plt.legend()
plt.grid(axis='y', alpha=0.3)
plt.show()
```

```
Extra Trees Residual Analysis
-----------------------------
Mean Residual: -0.3179
MAE: 0.5726
Maximum Absolute Error: 2.5119
Minimum Residual: -2.4457
Maximum Residual: 2.5119
```



Residual Analysis graph

7 My learnings

During this internship, I gained practical knowledge and hands-on experience in machine learning and industrial data analysis through the Quality Prediction in a Mining Process project.

The major learnings from this project are:

1. **Data Preprocessing:** Learned how to clean and prepare large industrial datasets using Pandas and NumPy.
2. **Exploratory Data Analysis:** Gained experience in analyzing industrial process parameters and understanding data distributions through visualizations.
3. **Feature Analysis:** Learned how to identify important process parameters and understand their contribution to model predictions.
4. **Machine Learning:** Gained practical experience in implementing Random Forest and Extra Trees regression algorithms.
5. **Model Evaluation:** Understood how to evaluate regression models using MAE, RMSE, and R^2 score.
6. **Model Comparison:** Learned how to compare multiple machine learning algorithms and select a better-performing model based on validation results.
7. **Project Development:** Improved problem-solving, analytical thinking, coding, and technical documentation skills.

Overall, this internship helped me understand the practical application of machine learning in industrial processes and strengthened my interest in data science and artificial intelligence.

8 Future work scope

The Quality Prediction in a Mining Process project can be further improved and extended in several ways:

1. **Advanced Machine Learning Models:** Explore algorithms such as XGBoost, Gradient Boosting, and Neural Networks to improve prediction performance.
2. **Real-Time Data Integration:** Integrate live sensor data from mining equipment to support continuous quality prediction.
3. **Additional Process Parameters:** Include more operational parameters to improve the understanding of factors affecting silica concentrate quality.
4. **Feature Selection and Optimization:** Apply advanced feature selection and hyperparameter tuning techniques to improve model performance.
5. **Prediction Dashboard:** Develop a web-based dashboard to display process parameters, predicted silica concentrate percentage, and model performance.
6. **Anomaly Detection:** Identify unusual variations in process parameters that may affect the quality of the final concentrate.
7. **Industrial Deployment:** Extend the model for practical use in mining plants to support quality monitoring and process optimization.

These improvements can help transform the current machine learning model into a more comprehensive industrial quality prediction system.